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% --- Example 2 Parameters ---
A2 = [0.76 -0.11; -0.1 0.54];
S2 = [0.37 -0.21; 0.25 0.21];
R2 = [0.3 0; 0 0.5];
Ko2 = -1/3;
Kq2 = 1;

k_max = 100;
x2 = zeros(2, k_max);
x2(:,1) = [0.5; -0.3]; % Initial condition

for k = 1:k_max-1
    % New Noise for Example 2
    if k <= 50
        wk = [1.2*sin(k); -0.5*cos(k)];
    else
        wk = [0; 0];
    end

    yk = A2*x2(:,k) + R2*wk;

    % Apply the sector-bound logic (Quantization)
    fk = max(min(yk, 1), Ko2);

    x2(:,k+1) = fk + S2*wk;
end

% Plotting Example 2
figure;
plot(0:k_max-1, x2(1,:), 'r', 'LineWidth', 1.5); hold on;
plot(0:k_max-1, x2(2,:), 'b', 'LineWidth', 1.5);
grid on;
title('Example 2: State Trajectories (Ko = -1/3)');
xlabel('k'); ylabel('x(k)');
legend('x_1', 'x_2');

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